



Yash Rampuria
Mechanical Engineering
Indian Institute of Technology Bombay

210100175
B.Tech.
Gender: Male
DOB: 18/05/2003

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2025	
Intermediate	CBSE	Delhi Public School Bengaluru South	2021	98.20%
Matriculation	CBSE	Delhi Public School Bengaluru South	2019	95.30%

Pursuing a Minor in **Artificial Intelligence & Data Science** by C-MInDS and **Honours** in Mechanical Engineering

SCHOLASTIC ACHIEVEMENTS

- Holding **Department Rank 4** out of 200+ students | Achieved **10/10** SPI (perfect grade) in 5th semester ['24]
- Awarded **3 AP** grades in **Guidance and Control of UAVs**, Fluid Mechanics Lab and **Microprocessors and Controls** Lab, given to the **top 1%** for exemplary performance | JEE Main: **99.93%**ile | JEE Advanced **99.27%**ile ['24]

PUBLICATIONS

- Yash Rampuria, et al., "IIT Bombay Racing Driverless: **Autonomous Driving** Stack for Formula Student AI," 2024. **Published** on arXiv, for research and development in self-driving Perception, SLAM, Motion Planning and Controls
- Yash Rampuria, et al., "Electron beam melting of Inconel alloys for **aerospace** applications: A review of microstructure and mechanical properties," under review COPEN-13th Int'l Conference on Precision, Micro, Meso, Nano Eng.

PROFESSIONAL EXPERIENCE

Amazon | **Software Engineering Intern**

[May'24 - Jul'24]

Awarded manager's recommendation for a full-time **return offer** | Completely **refactored** the integration tests package

- Deployed **8+ approved** code reviews to reduce user latency in integration testing by **automating** test data generation
- Implemented a **regression** testing framework for transport accounting using **Java JSON** configurations and TestNG
- Utilized **AWS - SQS/SNS** and **DynamoDb** and presented an SOP to senior engineers for **16+** dataflows in pipelines

Axisrooms - Hotelogix | **Machine Learning Intern**

[May'23 - Jul'23]

Awarded a **Letter Of Recommendation** for outstanding work in developing a model for forecasting hotel occupancy

- Implemented a new model with **30%** lesser MAPE and optimized it by tuning parameters to get the lowest AIC score
- Achieved **81% accuracy** by scrutinizing XGBoost, ARIMA, RNNs (LSTM) and their KPIs to select the lowest MAPE

ENGINEERING EXPERIENCE | RESEARCH PROJECTS

Chief Autonomous Systems Officer | IIT Bombay Racing Driverless | **Prof. S Anand**

[Mar'22 - Present]

Leading the **Level 4 Autonomous Vehicle** division in a team of 100+ | Represented India at Formula Student AI, UK

Leadership & Impact	- Responsible for India's 1st EV racecar integrated with vision, SLAM, path planning and controls Int'l exposure UK: 3rd in Real World AI, Top 5 in Simulations, Autonomous design and Business - Won Ather's System Intelligence award for ADAS and JLR's competition for best self-driving RnD
Computer Vision & Robotics	- Built a ROS integrated system on Ubuntu with LiDAR - Mono/Stereo Camera - IMU sensor fusion - Reduced latency from 100ms to 7ms with a C++ TensorRT engine for YOLOv5 object detection - Achieved 22% higher accuracy in depth estimation with a novel curvefit and Kalman Filter SLAM - Optimized a ResNet model for feature-point extraction by implementing parallel dataflow on CUDA - Reduced cross-track error 16% by non-linear pure pursuit control and simulated 60+ scenarios - Improved perception mAP from 0.91 to 0.98 by training with augmented data, noise & occlusions

Deep Learning for Cancer Cell Detection | MeDAL Lab | **Prof. Amit Sethi**

[Jul'23-Present]

Development and training of a custom segmentation model by integrating Vision Transformers with U-Net architectures

- Improved segmentation IoU by **19%**, applying a distance threshold in mask generation to distinguish crowded nuclei
- Implemented a novel cross-framework **architecture fusion** combining SOTA **PyTorch** CellViT and **Keras** StarDist
- Achieved **0.91** recall by finetuning for **47k+** nuclei and SOTA comparable results for custom model on PanNuke data

Aerial Robotics in Drones | INDUS Lab | **Bachelor Thesis, Prof. Vivek Sangwan**

[Dec'23 - Present]

Development and **control** of an autonomous multi-rotor aerial vehicle with a linked manipulator using **VICON** cameras

- Achieved < **2%** steady-state error and < **5s** settling time using precise **PID** tuning for hover and helical trajectories
- Optimized controller gains via **MATLAB / Gazebo** simulations for outerloop control with **10x** faster attitude innerloop
- Utilizing **feedback linearization** and low pass filters for aerial manipulation keeping stability during object interaction

Full stack aerial robotics project to design **path planning** and **autopilot** algorithms for unmanned fixed wing aircrafts

- Built a Simulink model to implement **6-DOF** dynamics of Cessna 620 with wind disturbances and actuator saturation
- Implemented rapidly exploring random tree (**RRT**) and **Voronoi** graphs for path planning with obstacle avoidance
- Achieved airspeed, altitude, and orientation control with < **3%** error for **20%** disturbance inputs with 50s period

AI - IoT Hackathon | **Indo-Japan Industry Summit** | **AWL Japan, Nedo Japan & IITBRP** [Jan'23 - Feb'23]

Supermarket digitization competition for groundbreaking **AI** solutions to locate misplaced items for swift rectification

- Awarded the **1st Prize** worth **60k** for on-site research of 3 stores, best business model and computer vision strategy
- Proposed **25%** higher profits by devising a solution to detect and classify misplaced items using **YOLOv5 - edge AI**
- Showcased the framework leveraging use of **RFID** alerts and monitoring systems to a panel of expert AI industrialists

Optimization of Multimodal Transportation | **IEOR Course Project** | **Prof. A Bhardwaj** [Jan'24 - May'24]

Technical stack for optimizing cost, time and distance between two geographical locations in a **graph based network**

- Formulated the decision variables and constraints on **Gurobi** to find the best possible path in a network of **50+** nodes
- Developed a **one-click solution** to ease transportation linking together government interests and customer needs

Texture Synthesis in Image Processing | **Course Project** | **Prof. Ajit V Rajwade** [Jun'23 - Dec'23]

Image processing project on texture synthesis with application in image hole filling, extrapolation and pattern generation

- Introduced an **improvement** to random sampling by synthesizing pixels of least SSD with a kernel around the seed
- Demonstrated an optimized kernel size of **21** pixels to for visually appealing results with computational efficiency

Road Accident Prevention Using Computer Vision | **Season of Code '22** | **WnCC, IITB** [Apr'22 - Jun'22]

Deep Learning / Convolutional Neural Network project to improve **road safety** by detecting and alerting sleepy drivers

- Implemented a model using **Keras** which alerts the driver when eyes close with confidence > **0.8** for a certain time
- Built the CNN with **Relu** activation and Categorical **Cross-Entropy** loss to classify between open and closed eyes

POSITIONS OF RESPONSIBILITY

Institute Student Mentor & Department Academic Mentor | **SMP, IIT Bombay** [Jun'23 - Present]

Selected out of **450+** students as a part of a mentorship team through an extensive process of interviews/peer reviews

- Mentoring **12** freshmen & **6** sophomores, providing counsel, ensuring academic well being and holding help sessions
- Helped assuage Branch-Change related concerns in Sophomore - 101, an event of importance to **1400+** students
- Acting as the first point of contact aiding the **communication** between the faculty, institute functionaries and students

Events and Public Relations Coordinator | **Entrepreneurship-Cell, IIT Bombay** [Jun'22 - Jan'23]

Asia's largest entrepreneurship promoting student body recognised by **NEN** patronage, **UNESCO** with **90+** members

- Ideated publicity campaigns for Eureka!, **Asia's largest** Business-model competition with prizes worth **INR 8 million+**
- Provided mentorship to **150+** startups by **30+** VCs and investors by strategizing Founders Garage and **Startup Expo**

TECHNICAL SKILLS

- **Programming Languages** : Python, C++, Java, JSON, HTML, $\text{\LaTeX}$, JavaScript, Excel
- **Frameworks** : ROS, GitHub, CUDA, Brazil, AWS, mavros, CRUX, JUnit, Mockito, Pixhawk
- **Packages/Libraries** : PyTorch, Pandas, TensorRT, Keras, Numpy, OpenCV, PyGame, SciPy

RELEVANT COURSES UNDERTAKEN

Robotics	Guidance and Control of UAVs, Microprocessors and Automatic controls, Mechatronic Systems
ML & Computer Vision	Digital Image Processing, Satellite Image Processing, Statistical Machine Learning and Data Mining, Computer Programming, Programming for Data Science, Intro to Machine Learning

EXTRA-CURRICULAR ACTIVITIES

Entrepreneurship	• National winner: in Indian Case Competition 5500+ participants ADAS startup pitch • Ranked among top 3 teams of BCG Ideathon out of 40+ students for a business pitch • Pitched a self made Business Model Canvas in EnPro, an entrepreneurship bootcamp
Debating	• Achieved 2nd position in Dips - Model United Nations '18 in the Human Rights Committee
Film Making and Acting	• Awarded the 1st runners up out of 30+ teams in The Times of India dramatics competition • Attained 50k+ views on Instagram for filming the Mood Indigo Insti Nite aftermovie • Secured 1st prize against 1.4k+ students in Freshiezza Spoof making and 3rd in Ad making